Week 3 Internship Report: Linux Logs, Sysmon, Auditd & Snort

SOC Analyst Internship Performance & Technical Summary

Field	Details
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Domain	SOC Analyst
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Report Status	Completed

1. Executive Summary

This report covers the theoretical concepts and practical hands-on tasks completed during Week 3 of the SOC Analyst internship. The focus was on deep endpoint telemetry (Windows Sysmon and Linux `auditd`), network intrusion detection with Snort, and understanding log ingestion pipelines. Additionally, multiple TryHackMe and LetsDefend labs were completed to reinforce these core security operation capabilities.

2. Theoretical Understanding

2.1 Endpoint Detection and Response (EDR)

- **Definition:** Security solution that continuously monitors endpoints for malicious behavior and provides automated or guided response capabilities.
- **Importance:** Provides rich telemetry (process execution trees, detailed command lines) crucial for investigating modern attacks that bypass traditional antivirus controls.
- **EDR vs. Antivirus (AV):** AV is signature-based and primarily focuses on static file prevention; EDR is behavior-based and focuses on continuous visibility, threat detection, investigation, and incident response.

2.2 Sysmon vs. Standard Windows Event Logs

- **Sysmon:** Provides deep, actionable security visibility across key operational activities:
  - `Event ID 1`: Process Creation (includes full command-line arguments and process hashes)
  - `Event ID 3`: Network Connections (maps network activity to process IDs)
  - `Event ID 11`: File Creation (tracks dropped binaries and scripts)
  - `Event ID 22`: DNS Queries (exposes domain resolution attempts)
- **Standard Windows Logs:** Security, System, and Application logs are limited in default configurations and lack the granular context required for advanced threat hunting and process genealogy analysis.

2.3 Linux auditd vs. Syslog

- **Linux auditd:** Performs kernel-level auditing with highly structured output. Captures specific system calls (e.g., `EXECVE`) and is tailored for security auditing and compliance compliance monitoring.
- **Syslog:** Application-level message logging service utilizing free-text formats, primarily designed for general system troubleshooting and operational monitoring.

2.4 Snort Intrusion Detection & Prevention System (IDS/IPS)

- **IDS Mode (Passive):** Monitors network traffic passively and generates alert events without interfering with packet flow.
- **IPS Mode (Inline):** Sits directly in the network traffic path to actively drop or block malicious packets in real time.
- **MITRE ATT&CK Mapping:** Snort rules are mapped to specific techniques to measure and validate defensive detection coverage.

3. Lab Writeups Summary

3.1 TryHackMe: Linux Logging for SOC

- Explored system log locations including `/var/log/auth.log`, `syslog`, and `auditd` outputs.
- Mastered command-line audit investigation utilities such as `ausearch` and `aureport`.
- *Key Takeaway:* Linux logs are unstructured and vast, but highly detailed when queried with targeted filters.

3.2 TryHackMe: Intro to Logs & Log Analysis

- Analyzed common log formats (JSON, CSV, Combined Log Format) and standardized logging structures.
- Practiced parsing raw log files using core CLI utilities (`awk`, `grep`, `sed`, `sort`, `uniq`).
- Leveraged CyberChef for data decoding and regular expressions (Regex) for field extraction.

3.3 TryHackMe: Snort

- Mastered core Snort operational modes: Sniffer (`-v`), Logger (`-l`), and NIDS/NIPS (`-c`).
- Gained experience crafting custom rules and parsing raw packet captures (`.pcap`).

3.4 LetsDefend: SOC Lessons (SIEM 101, Incident Management)

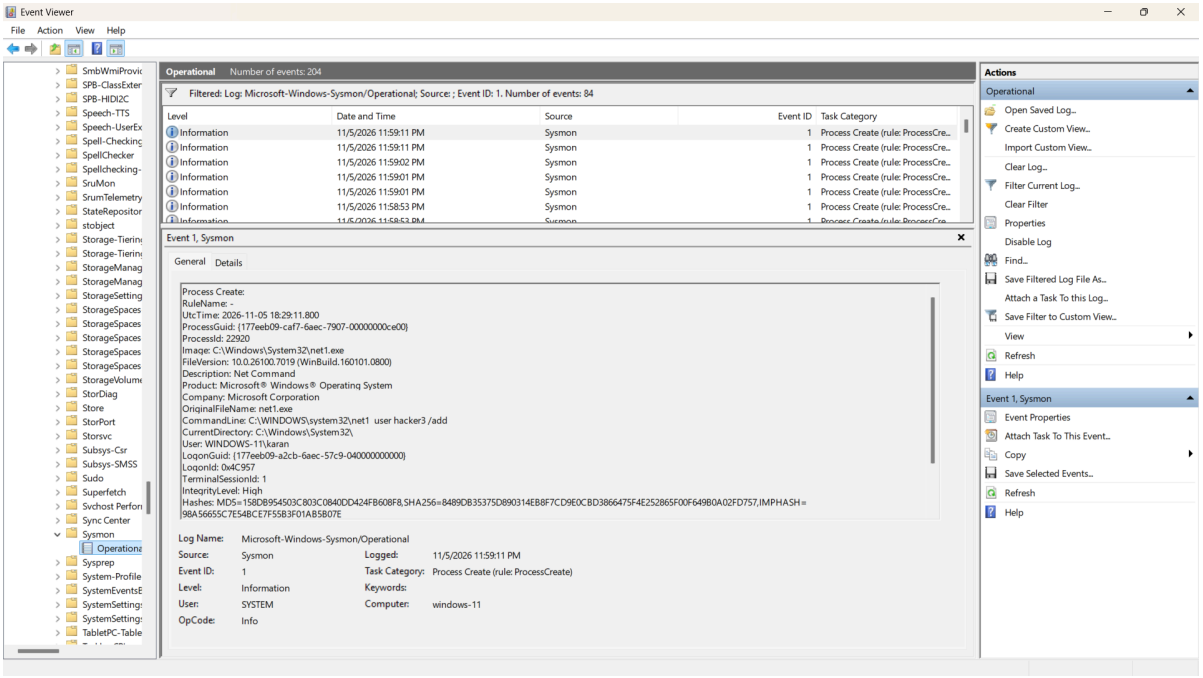
- Studied the Incident Response lifecycle based on the NIST framework.
- Understood SIEM telemetry pipelines: log ingestion, normalization, parsing, and alert triage.
- Practiced investigating SIEM alerts and evaluating context to filter out false positives.

4. Hands-On Practical Tasks

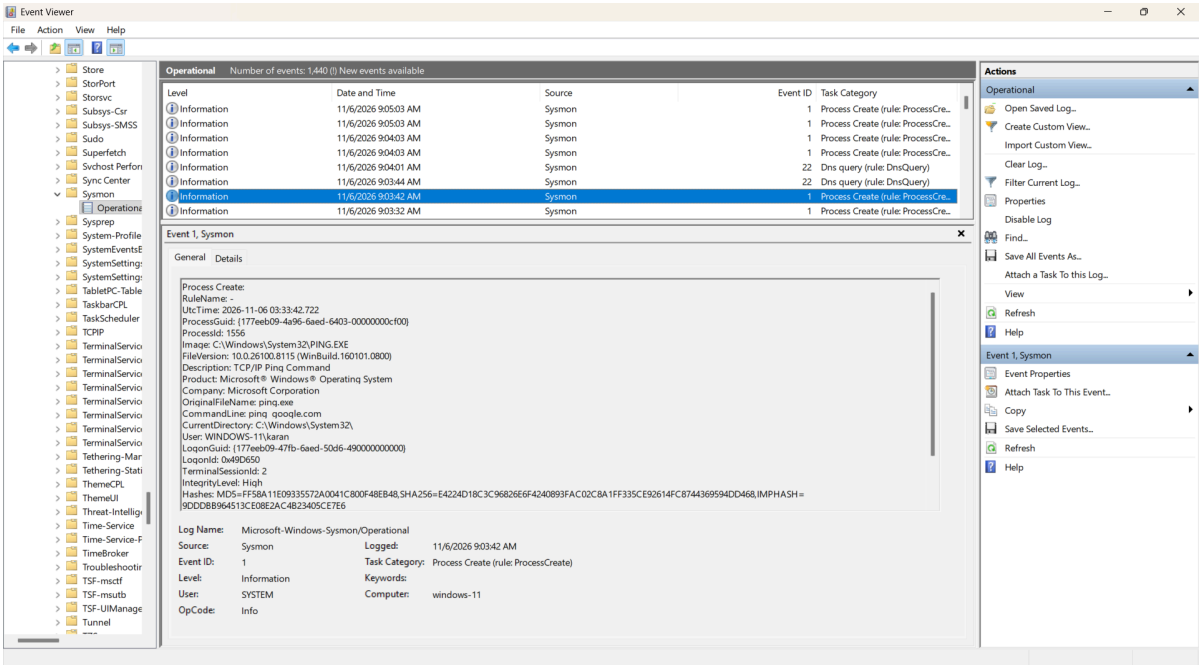
4.1 Tasks 1 & 2: Sysmon Installation & Event Generation

- Installed Windows Sysmon using the SwiftOnSecurity custom configuration XML.
- Generated and investigated high-fidelity telemetry events:
    - **Event ID 1:** Command-line execution via PowerShell and `net user`.
    - **Event ID 3:** Network connection tracking via ICMP `ping` sweeps.
    - **Event ID 11:** Dropped payload file creation events.

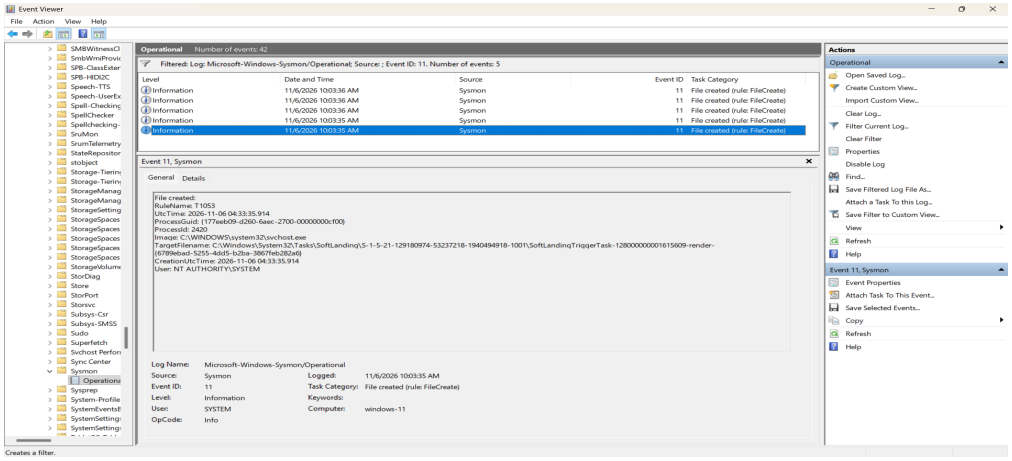
Evidence: Sysmon Event Viewer showing process creation telemetry for `Event ID 1`.



- **Event ID 3**



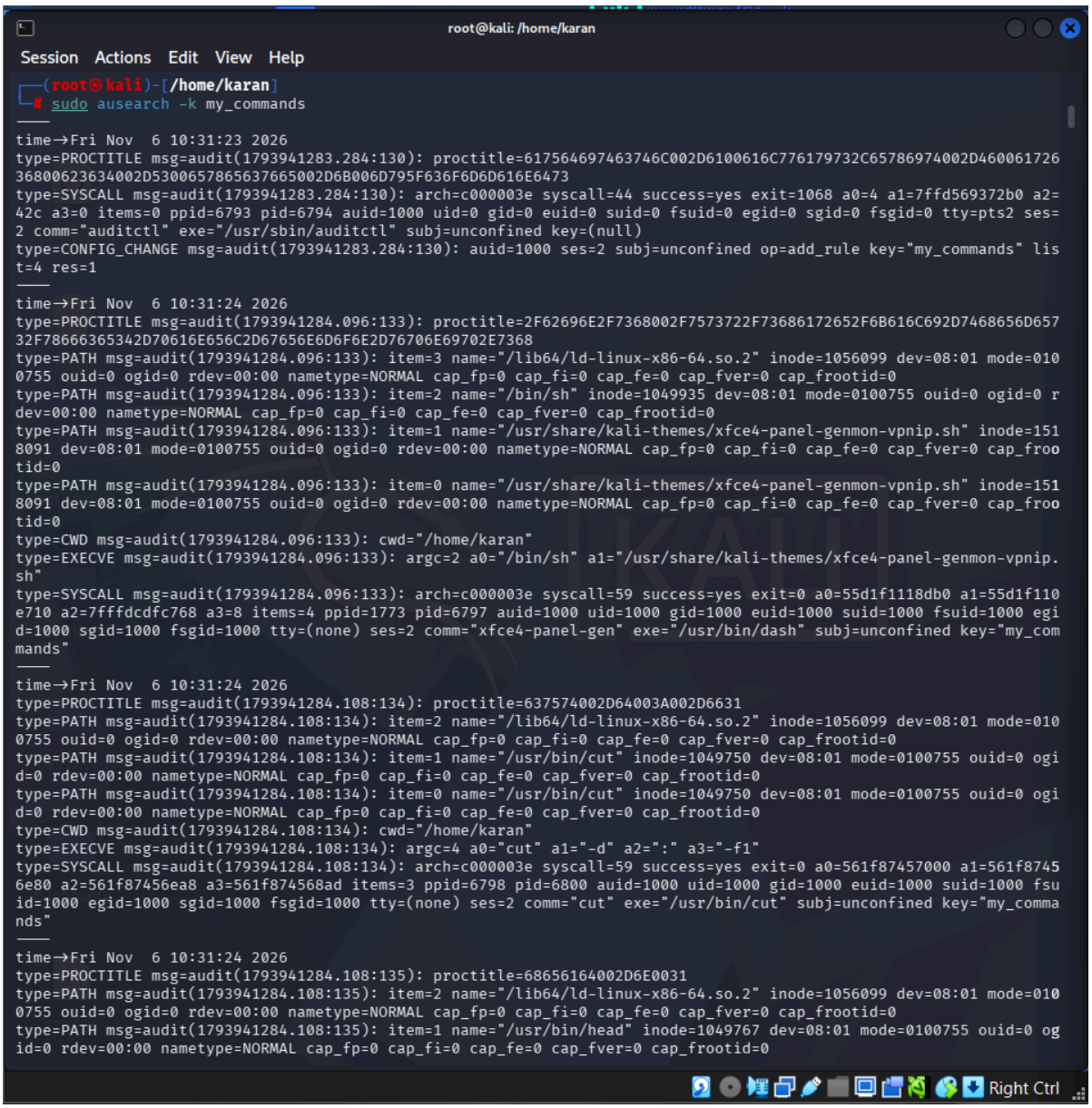
- **Event ID 11**



4.2 Task 3: Linux Auditd Setup

- Installed and enabled `auditd` on Kali Linux.
- Configured custom audit rules to monitor executive system calls (`execve`).
- Queried logged command executions using `ausearch -k my_commands`.

Evidence: Terminal output showing `ausearch` results for `EXECVE` system call telemetry.



4.3 Tasks 4 & 5: Snort Setup & Advanced Rule Configuration

- Installed Snort and configured the engine within `local.rules`.
- Created baseline detection rules for ICMP, SSH, and HTTP traffic.
- Engineered an advanced detection rule for Nmap SYN port scans mapped to MITRE ATT&CK Technique T1046 (Network Service Discovery).
- *Environment Note:* Documented and resolved an environment-specific syntax issue during rule validation in Kali Linux.

5. Telemetry Cheat Sheet

Event ID / Rule	Source	Signature / Meaning	Operational Use Case
4624	Windows Security	Successful User Account Logon	Investigating normal vs. anomalous user authentication
4625	Windows Security	Failed User Account Logon	Detecting brute-force or password spraying attempts
4688	Windows Security	Process Creation	Tracking binary execution across endpoints
4720	Windows Security	Local User Account Created	Detecting unauthorized account creation (T1136)
1102	Windows Security	Audit log cleared	Detecting anti-forensics. Attackers clear logs to hide their tracks; this is a high-severity alert.
7045	Windows Security	New service installed	Detecting persistence. Malware often installs itself as a Windows service to survive reboots.
1	Sysmon	Process Create (with CLI)	Monitoring process trees, hashes, and arguments
3	Sysmon	Network Connection	Detecting C2 communication or lateral movement
11	Sysmon	File Create	Detecting dropped malware payloads and scripts
22	Sysmon	DNS Query	Uncovering DNS-based C2 or suspicious domain lookups
8	Sysmon	CreateRemoteThread	Detecting process injection. One process starting a thread

Event ID / Rule	Source	Signature / Meaning	Operational Use Case
			in another process (e.g., malware injecting into svchost).
EXECVE	Linux auditd	System Command Execution	Auditing command execution on Linux endpoints
USER_LOGIN	Linux auditd	User Authentication	Tracking user login sessions on Linux systems
USER_AUTH	Linux auditd	Authentication Attempt	Detecting failed login attempts and privilege escalation
1000001	Snort	ICMP Ping Detected	Identifying network reconnaissance sweeps
1000002	Snort	SSH Connection Attempt	Monitoring remote administrative connection attempts
1000003	Snort	Unencrypted HTTP Traffic	Auditing web service interaction and plain-text payloads
1000004	Snort	Nmap SYN Scan Signature	Detecting active port scanning and host discovery (T1046)

6. Conclusion

Week 3 provided an in-depth exploration of endpoint and network security telemetry. Combining Windows Sysmon, Linux `auditd`, and Snort established a comprehensive foundation for detecting, investigating, and responding to adversarial tactics across diverse operational environments.

The practical exercises strengthened core analysis skills, and troubleshooting environment-specific Snort configuration issues offered valuable experience in managing real-world security tool deployments.